

THE TWENTY-EIGHTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION

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A-1. Let $f(x) = a_1 \sin x + a_2 \sin 2x + \cdots + a_n \sin nx$, where $a_1, a_2, \dots, a_n$ are real numbers and where n is a positive integer. Given that $|f(x)| \leq |\sin x|$ for all real x , prove that

$$|a_1 + 2a_2 + \cdots + na_n| \leq 1.$$

A-2. Define S_0 to be 1. For $n \geq 1$, let S_n be the number of $n \times n$ matrices whose elements are nonnegative integers with the property that $a_{ij} = a_{ji}$, $(i, j = 1, 2, \dots, n)$ and where $\sum_{i=1}^n a_{ij} = 1$, $(j = 1, 2, \dots, n)$. Prove

(a) $S_{n+1} = S_n + nS_{n-1}$,
(b) $\sum_{n=0}^{\infty} S_n \frac{x^n}{n!} = \exp(x + x^2/2)$, where $\exp(x) = e^x$.

A-3. Consider polynomial forms $ax^2 - bx + c$ with integer coefficients which have two distinct zeros in the open interval $0 < x < 1$. Exhibit with a proof the least positive integer value of a for which such a polynomial exists.

A-4. Show that if $\lambda > \frac{1}{2}$ there does not exist a real-valued function u such that for all x in the closed interval $0 \leq x \leq 1$, $u(x) = 1 + \lambda \int_x^1 u(y) u(y-x) dy$.

A-5. Show that in a convex region in the plane whose boundary contains at most a finite number of straight line segments and whose area is greater than $\pi/4$ there is at least one pair of points a unit distance apart.

A-6. Given real numbers $\{a_i\}$ and $\{b_i\}$, $(i = 1, 2, 3, 4)$, such that $a_1b_2 - a_2b_1 \neq 0$. Consider the set of all solutions (x_1, x_2, x_3, x_4) of the simultaneous equations

$$a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 = 0 \text{ and } b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 = 0,$$

for which no x_i ($i = 1, 2, 3, 4$) is zero. Each such solution generates a 4-tuple of plus and minus signs (signum x_1 , signum x_2 , signum x_3 , signum x_4).

- (a) Determine, with a proof, the maximum number of distinct 4-tuples possible.
(b) Investigate necessary and sufficient conditions on the real numbers $\{a_i\}$ and $\{b_i\}$ such that the above maximum number of 4-tuples is attained.

B-1. Let $(ABCDEF)$ be a hexagon inscribed in a circle of radius r . Show that if $\overline{AB} = \overline{CD} = \overline{EF} = r$, then the midpoints of $\overline{BC}$, $\overline{DE}$, $\overline{FA}$ are the vertices of an equilateral triangle.

B-2. Let $0 \leq p \leq 1$ and $0 \leq r \leq 1$ and consider the identities

(a) $(px + (1-p)y)^2 = Ax^2 + Bxy + Cy^2$,
(b) $(px + (1-p)y)(rx + (1-r)y) = \alpha x^2 + \beta xy + \gamma y^2$.

Show that (with respect to p and r)

(a) $\max\{A, B, C\} \geq 4/9$,
(b) $\max\{\alpha, \beta, \gamma\} \geq 4/9$.

B-3. If f and g are continuous and periodic functions with period 1 on the real line, then $\lim_{n \rightarrow \infty} \int_0^1 f(x) g(nx) dx = (\int_0^1 f(x) dx)(\int_0^1 g(x) dx)$.

B-4. (a) A certain locker room contains n lockers numbered $1, 2, 3, \dots, n$ and all are originally locked. An attendant performs a sequence of operations $T_1, T_2, \dots, T_n$ whereby with the operation $T_k, 1 \leq k \leq n$, the condition of being locked or unlocked is changed for all those lockers and only those lockers whose numbers are multiples of k . After all the n operations have been performed it is observed that all lockers whose numbers are perfect squares (and only those lockers) are now open or unlocked. Prove this mathematically.

(b) Investigate in a meaningful mathematical way a procedure or set of operations similar to those above which will produce the set of cubes, or the set of numbers of the form $2m^2$, or the set of numbers of the form $m^2 + 1$, or some nontrivial similar set of your own selection.

B-5. Show that the sum of the first n terms in the binomial expansion of $(2 - 1)^{-n}$ is $\frac{1}{2}$, where n is a positive integer.

B-6. Let f be a real-valued function having partial derivatives and which is defined for $x^2 + y^2 \leq 1$ and is such that $|f(x, y)| \leq 1$. Show that there exists a point (x_0, y_0) in the interior of the unit circle such that

$$\left(\frac{\partial f}{\partial x}(x_0, y_0)\right)^2 + \left(\frac{\partial f}{\partial y}(x_0, y_0)\right)^2 \leq 16.$$